

Jonathan Tadesse labb 2

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1 Problem 1

1.1 Experimenting with different separation distances

Based on the theory of interference from two point sources with the same wavelength, we can either have **constructive interference** or **destructive interference** for two waves arriving at a point.

Let λ be the wavelength for the two point sources. We have **constructive interference** at a point if the path difference Δr between the two waves is an integer multiple of the wavelength:

$$\Delta r = m\lambda, \quad \text{for } m = 0, \pm 1, \pm 2, \dots$$

where m is an integer. The path difference Δr is approximated as $d \sin \theta$ where d is the separation between the sources and θ is the angle relative to the normal to the midpoint between the sources.

For **destructive interference**, the condition is:

$$\Delta r = \left(m + \frac{1}{2}\right) \lambda, \quad \text{for } m = 0, \pm 1, \pm 2, \dots$$

In our experiment, with $\lambda = 5$, plotting the wave interference for the following separation distances $[10, 17, 25, 40, 84]$ yielded the following plots.

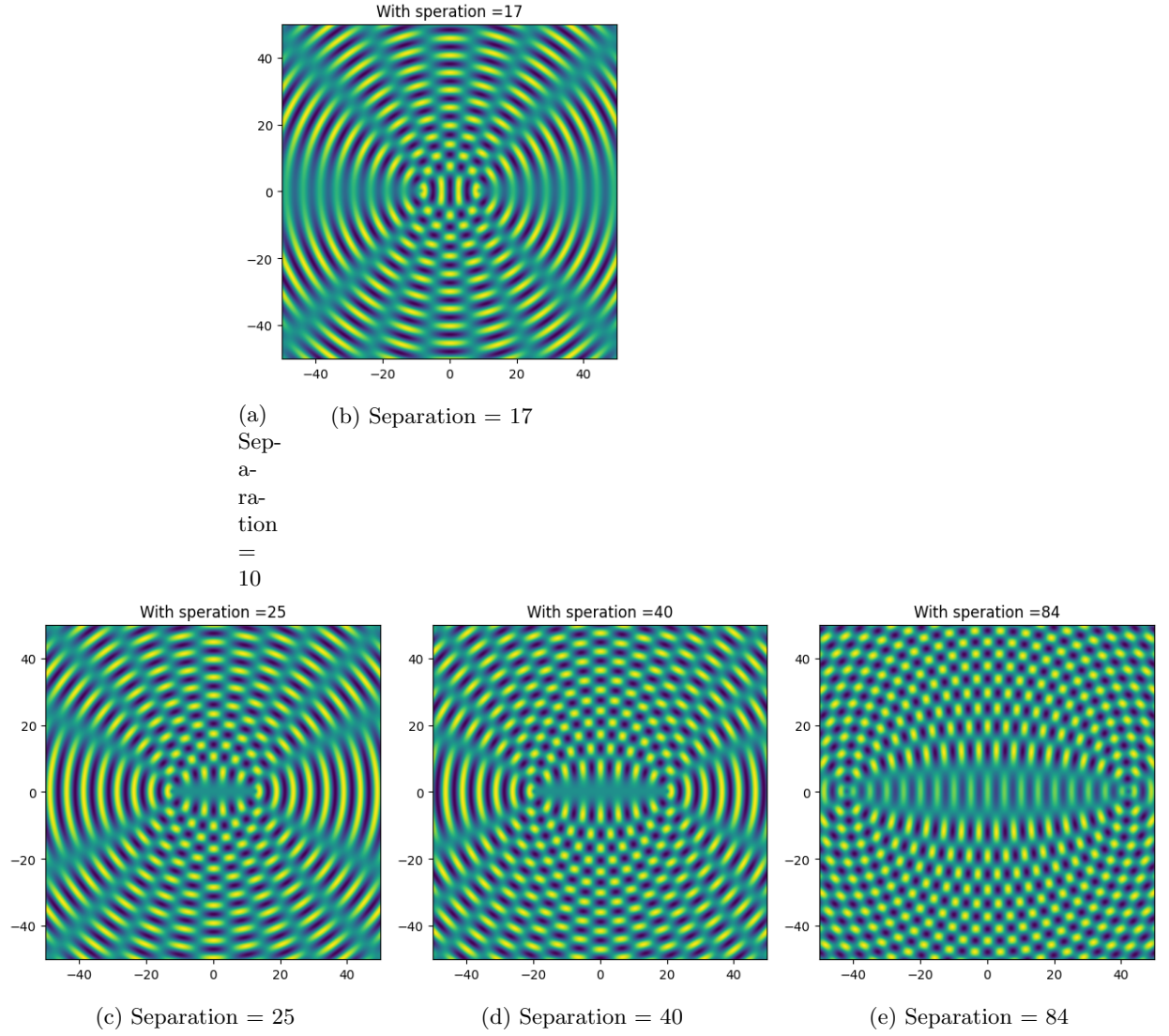


Figure 1: Interference patterns for different source separations

We noticed that in the plots with smaller separation distances less than 20, the nodes (points of destructive interference) with stream of blurry blue lines were generally larger. This was also observed for the antinodes (points of constructive interference) however, they were generally larger than the nodes. As the separation distance grew, we observed that the nodes, i.e., the streams of blurry blue lines, shrank and were largely confined to the region between the two point sources along the x -axis as in figure e.

1.2 Adding another point source

We added another point source with the same wavelength and amplitude and observed that the results differed depending on the separation. For the case when the separation $d = 10$ we observed a big region of constructive wave interference radiating along $\pm x$ -axis and a smaller one along the $\pm y$ -axis. We also observed that a significant portion of the waves cancelled out between the above streams. When the separation was $d = 17$ nearly all of the waves had cancelled out except for a few streams, this is likely because the path difference at the point of destructive interference is not an integer multiple of the wavelength $\lambda = 5$. As the separation grew we observed more radial symmetry and regular regions of destructive and constructive wave interference. For example in the case $d = 84$ we see the formation of concentric circles with regions of destructive interference in between them.

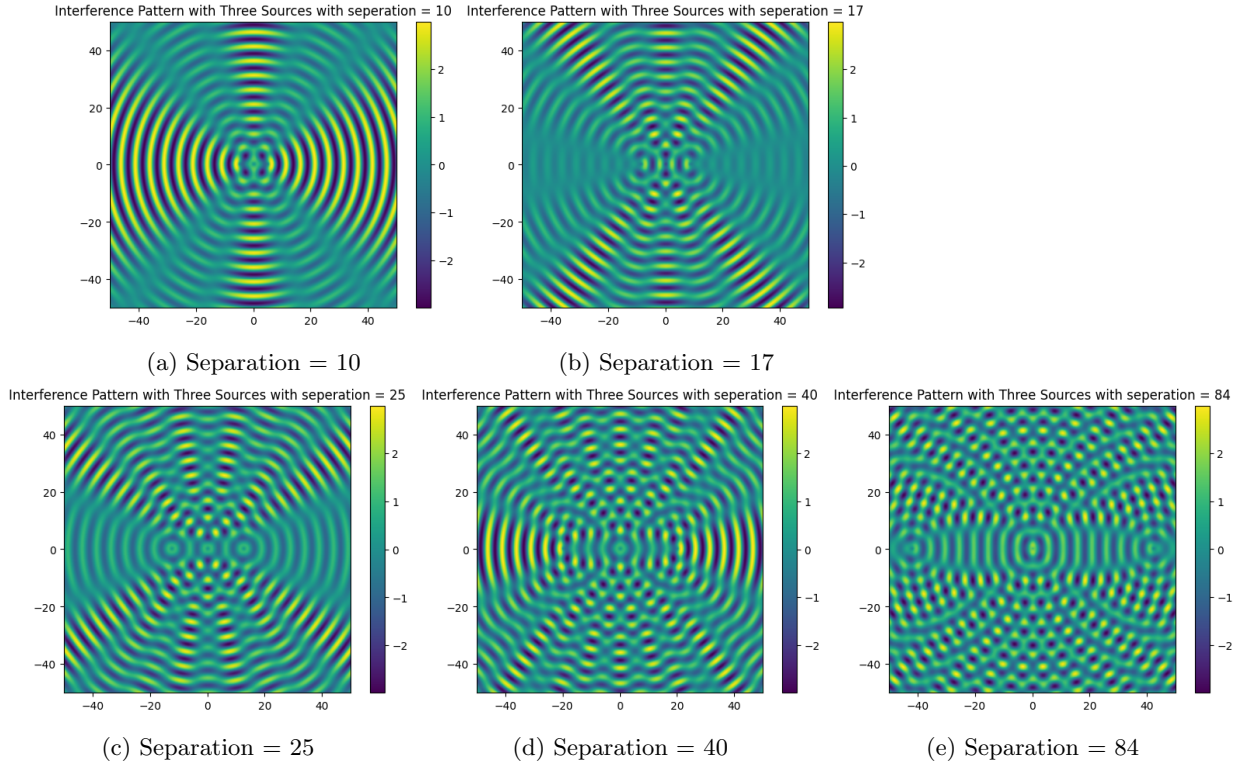


Figure 2: Interference patterns for 3 different sources with different separations

2 Problem 2

The distance between two interference maxima is given by the formula $\Delta x = \lambda D/d$ with a usual maxima when $\Delta x = 0$ i.e at the midpoint of the screen

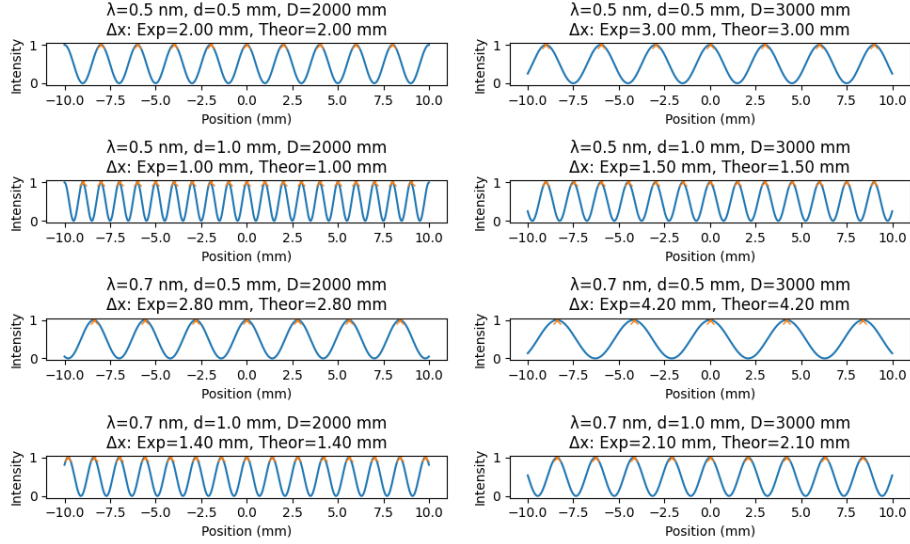
detector. We simulated the double slit experiment with the following different values

$d : [0.5, 1.0]$,

$\lambda : [0.0005, 0.0007]$

$D : [2000, 3000]$.

We then calculated the theoretical distance between two interference maxima and then compared it to a numerical approximation. The following plot figure 2 displays the interference pattern for different values of λ , D and d . As expected Δx grew as the wavelength got bigger, this is sound since λ is at the numerator. As d got bigger we observed a smaller Δx and vice versa which is also sound. The greatest Δx was obtained with the smallest value of d and and the greatest values of λ and D i.e $[d = 0.5, \lambda = 0.7, D = 3000]$. The plot showed that the numerical approximation to Δx matched the theoretical calculation up to three significant digits using the inbuilt `np.peaks` method from `numpy` library.



(a) Double slit experiment for different values

Figure 3

3 Problem 3

Now we model a more realistic single slit experiment with uniform distance between the sources. We investigate how the intensity peak varies with w and N (number of source points). We theorized that the width of the central maxima is inverse proportional to the slit width.

$$\theta \approx \frac{\lambda}{w}$$

. where λ is the wavelength, in our case 0.00005. This implies that, increasing the slit width would result in a narrower central maxima and thus a higher intensity. Similarly, decreasing the slit width would increase the width of the central maxima i.e smaller intensity. We choose to vary N and w independently to see what results we get, figure 4 was simulated with $N = 2, 3, 5, 10$ and $w = 0.05$.

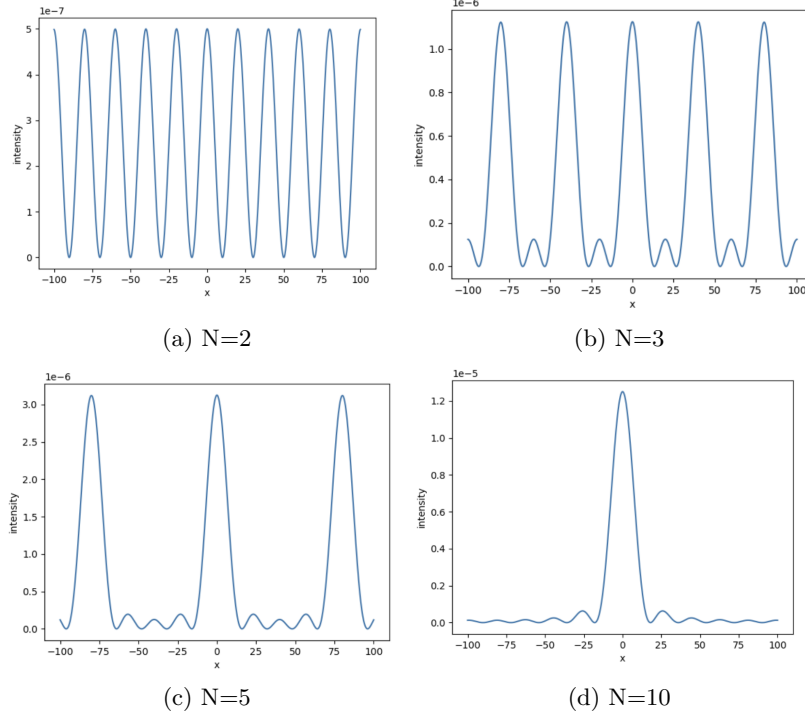


Figure 4

What can be observed from figure 4 is that with greater N the distance between the peaks get bigger, it can also be observed that greater N implies a higher intensity peak. For instance, for $N = 2$ we have the maximum peak at $5 \cdot 10^{-7}$ whereas for $N = 10$ we have the maximum peak at $1.2 \cdot 10^{-5}$.

We then simulated the experiment with $N = 5$ sources and varied w between 0.025 and 0.1, figure 5 shows the result.

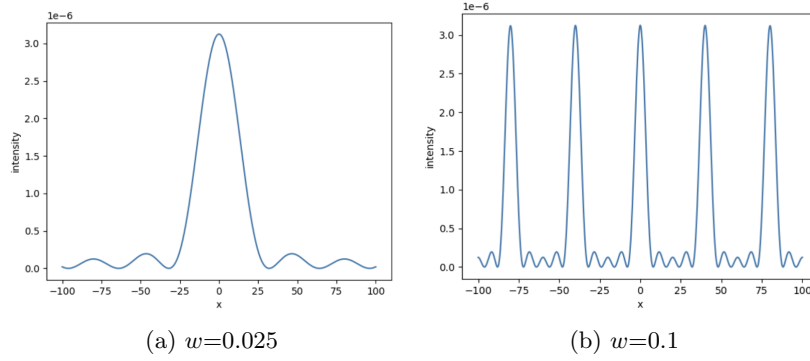


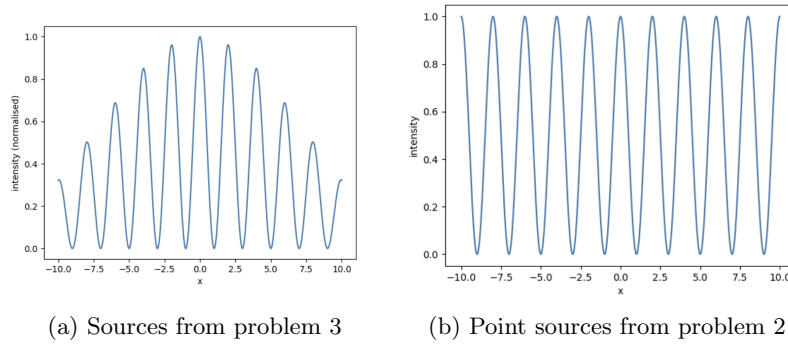
Figure 5

The figures seem to indicate that a smaller w resulted in a wider peak, additionally the central maxima seem to dominate the most and the surrounding maximas are almost unnoticeable. With greater w we see in plot (b) that the maximas seem to be more uniform and with the same intensity peak.

4 Realistic double slit

We conducted a more realistic double-slit experiment using the same slits from problem 3, with $2N$ sources, slit distance $d = 0.5$, $N = 10$, and $w = 0.05$.

Figure (a) shows the result.



What can be observed with the graph on problem 2 and (b) is that they seem to agree on the whole, they have similar distances between their peaks.